Statistical Techniques (7MA505)
Tutorial - Week 3
16 October 2020

Examples of confidence interval calculations for standard normal and Student's (Gossert's) t-distribution

Area under the Normal Distribution Curve

1. We noted that approximately 95% of the area under the normal curve lies in the interval given by $\mu \pm 2\sigma$.
2. For $\mu = 0, \sigma = 1$, the standard normal distribution, to determine what lies under 95% of the curve, we note the following: The upper interval limit is at at the quantile 97.5. The lower interval limit is at the quantile 2.5. So:
3. `qnorm(0.975)` gives: 1.959964. `qnorm(0.025)` gives: -1.959964. Note that these are identical values, just one being positive and the other negative. (Explain this.) So we have these values as approximately $[-2, 2]$.

Area under the Normal Distribution Curve (contd.)

1. Now let's look at the situation for 99.7% of the normal curve's area.
2. We noted that as being the area spanning the interval $\mu \pm 3\sigma$.
3. This gives us quantile limits of 0.15%, 99.85%. (Explain this.)
4. We get the following then:
5. `qnorm(0.9985); qnorm(0.0015)` giving $2.967738 - 2.967738$.
6. This is (approximately) what we want to show: The interval $[-3\sigma, 3\sigma]$.

Area under the Normal Distribution Curve (contd.)

1. Now let's look at the situation for 68% of the normal curve's area.
2. Show that that area covers $\pm 1\sigma$ from the mean, μ.

Test for A Data Set Being Normally Distributed

```
# To look at lots of data sets available in R:
data()
# We will take one data set, relating to
# Girth, Height and Volume for Black Cherry Trees
data(trees)
dim(trees)    # Dimensions of the trees array
# We choose column 1 of this data as: trees[,1]
# We choose column 2 of this data as: trees[,2]
# We choose column 3 of this data as: trees[,3]
# For each column, look at their histograms, look at a
# QQ plot, and use the Shapiro-Wilk significance test of
# coming from a normal distribution.
```

Test for A Data Set Being Normally Distributed (contd.)

```
hist(trees[,1])
qqnorm(trees[,1])
qqline(trees[,1])
# Draws a line through the plot of quantiles in our data
# versus those of the normal distribution.  Does the line
# fit the data (very) well?
# Shapiro-Wilk test:
shapiro.test(trees[,1])
# Null hypothesis of this test: data normally distributed.
# If p-value is less than the chosen significance (alpha)
# level, very often 0.05, then the null hypothesis is rejected.

# Box plot (in the lecture, trees[,3] was used):
boxplot(trees[,3])
```

Example from Lecture

- A random sample of 80 tablets had an average potency of 15mg. Assume σ is known to be 4mg.
- $\bar{x} = 15, \sigma = 4, n = 80$
- A 95% confidence interval for $\bar{x}$ is:

$$15 \pm 1.96 \times \frac{4}{\sqrt{80}} = \{14.12, 15.88\}$$

- Doing this in R:
- `qnorm(0.025, mean=15, sd=4/sqrt(80))`
- That gives: 14.12348
- `qnorm(0.975, mean=15, sd=4/sqrt(80))`
- That gives: 15.87652

A Few Questions Where the Population Standard Deviation is Known

- Standard normal distribution, z-score transformation: $Z = \frac{x-\mu}{\sigma}$, of mean 0, standard deviation 1.
- 95% confidence interval for estimate of population mean, μ, by $\bar{x}$ if σ is known is given by:

$$\bar{x} \pm 1.96 \times \frac{\sigma}{\sqrt{n}}$$

- Mean of the sampling distribution of $\bar{x}$, $\mu_{\bar{x}} = \mu$, the population mean. Standard deviation of the sampling distribution, $\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$. This is estimated by the standard error of $\bar{x}$, $\mathrm{SE}_{\bar{x}} = \frac{s}{\sqrt{n}}$.
- Understand well the assumptions in regard to normality.

1. 95% confidence interval for estimate of population mean, μ, by $\bar{x}$ if σ is known is given by:

$$\bar{x} \pm 1.96 \times \frac{\sigma}{\sqrt{n}}$$

2. The weight of eggs sold by the Highlands grocery store is known to be normally distributed with a standard deviation of 18 grams. A quality control of eggs sold by this grocery store yields the average weight of 42 grams for a sample of twenty eggs. Determine the 95% confidence interval for the expected weight of eggs sold by this grocery store.

Work Through This Example (Included Also in Lecture 3)

```
set.seed(3142)  # For reproducibility of random numbers.
x <- round(100*rnorm(25)) + 300
# 25 normal values; then: rescaled, translated, rounded.
hist(x)                          # Check the histogram.
qqnorm(x)       # Check how normal the resulting data is.
# If we wished, we could instead just specify this data:
x2 <- c(238, 474, 359, 256, 481, 323, 151, 340, 284, 417,
424, 309, 139, 237, 269, 46, 350, 408, 223, 408, 286,
224, 238, 119, 343)

# Estimate mean salary and find 90% and
# 95% confidence intervals for the mean.

# Work first on the assumption that the mean is to be
# estimated and the standard deviation is 111.
# (Note this assumption here.)
```

Confidence Intervals when the Data is Normally Distributed (contd.)

- Looking for: Confidence Interval for the sample mean when the variance is known. A 95% confidence interval for $\bar{x}$ if σ is known is given by:

$$\bar{x} \pm 1.96 \times \frac{\sigma}{\sqrt{n}}$$

- Check this limit value of 1.96, for the normal distribution: `qnorm(0.975)` and (getting the negative of what we have just obtained), `qnorm(0.025)`.
- R gives us the following (which we can round to 3 significant digits): ± 1.959964
- How about the limits for a 90% confidence interval:
- Use `qnorm(0.95)` and find 1.644854

From lecture: 26 measurements of the potency of a single batch of tablets

```
x <- c(498.38, 489.31, 505.5, 495.24, 490.17, 483.2,
488.47, 497.71, 503.41, 482.25, 488.14, 492.22, 483.96,
473.93, 463.4, 493.65, 499.48, 496.05, 494.54, 508.58,
488.42, 463.68, 492.46, 489.45, 491.57, 489.33)

mean(x)
[1] 490.0962

sd(x)
[1] 10.78317
```

From lecture: 26 measurements (contd.)

```
t.test(x)

One Sample t-test
data:  x
t = 231.75, df = 25, p-value < 2.2e-16
alternative hypothesis: true mean is not equal to 0
95 percent confidence interval:
 485.7407 494.4516
sample estimates:
mean of x
 490.0962
```

- Population standard deviation, σ, not known, then use t-distribution.
- The 95% confidence interval when σ is unknown is:

$$\bar{x} \pm t_{0.025} \times \frac{s}{\sqrt{n}} \qquad \text{df} = n - 1$$

1. The weight of eggs sold by the Downlands grocery store is known to be normally distributed. A quality control of eggs sold by this grocery store yields the average weight of 42 grams and the empirical standard deviation of 18 grams for a sample of twenty eggs. Determine confidence intervals on the confidence levels 90%, 95% and 99% for the expected weight of eggs sold by this grocery store.

Quickly Work Through This Example from the Lecture

- (Verzani, p. 272.) A sample of 30 students have an average height of 66 inches with a standard deviation of 4 inches. Assume normally distributed. What is an 80% confidence interval for the mean height of all the population of students?
- From the lecture, we want: $\bar{x} \pm t^* \cdot \mathrm{SE}(\bar{x})$
- With $\sigma/\sqrt{n}$, the Standard Error of $\bar{x}$.
- So: `SE <- 4/sqrt(30)`
- 80% confidence interval: so we cover the interval that has 10% ignored at the lower end, and 10% ignored at the top end. So the top end limit point is $1 - (1.0 - 0.8)/2$
- (The bottom end limit point is the negative value, given by, not 0.9 but rather 0.1.)

Example (contd.)

Arising from the First part of Lecture 3 – On Confidence Intervals (contd.)

- Those confidence interval values from the distribution are given by: `qt( 1 - (1 - 0.8)/2, df = 29)`
- (Get: 1.311434 Try this also: `qt( 0.1, df = 29)`)
- For our given data, the confidence interval is then:
- `66 + SE * qt( 1 - (1-0.8)/2, df = 29)` and
- `66 - SE * qt( 1 - (1-0.8)/2, df = 29)`

Another Example – From the Lecture

- (Verzani, p. 273.) A barista aims at a 25-second espresso shot resulting in 2 ounces of espresso. This person wants to test that. Eight shots are poured and results measured: 1.95, 1.8, 2.1, 1.82, 1.75, 2.01, 1.83, 1.9 ounces. Find the 90% confidence interval for the mean. Does it include 2.0?
- (When we move to Hypothesis Testing, we will address this as a null hypothesis to be tested.)
- `ozs <- c( 1.95, 1.8, 2.1, 1.82, 1.75, 2.01,1.83, 1.9)`
- `SE <- sd(ozs)/sqrt(8)`
- SE is: 0.04187567
- What we want: $\bar{x} \pm t^* \cdot \mathrm{SE}(\bar{x})$
- (Continue this and consider the outcome.)

Work Through This Example from the Lecture.

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# Estimate mean salary and find 90% and
# 95% confidence intervals for the mean.

# Work first on the assumption that the mean is to be
# estimated and the standard deviation is 111.
# (Note this assumption here.)
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Confidence Intervals when the Data is Normally Distributed (contd.)

- Looking for: Confidence Interval for the sample mean when the variance is known. A 95% confidence interval for $\bar{x}$ if σ is known is given by:

$$\bar{x} \pm 1.96 \times \frac{\sigma}{\sqrt{n}}$$

- Check this limit value of 1.96, for the normal distribution: `qnorm(0.975)` and (getting the negative of what we have just obtained), `qnorm(0.025)`.
- R gives us the following (which we can round to 3 significant digits): ± 1.959964
- How about the limits for a 90% confidence interval:
- Use `qnorm(0.95)` and find 1.644854

Confidence Intervals when the Data is Normally Distributed (contd.)

- Then, for the confidence interval limits for the 95% interval:
- mean(x2) + qnorm(0.975) * (sd(x2)/sqrt(25))
- mean(x2) - qnorm(0.975) * (sd(x2)/sqrt(25))
- Get: [250.2606, 337.4194], better written as: [250.26, 337.42]
- For the 90% confidence interval limits, find:
- $(257.267, 330.413)$

Confidence Intervals when the Data is Normally Distributed (contd.)

- What if our knowledge of the standard deviation ($= 111$) was incorrect? That we do now know this, and hence that we must estimate it from the given data.
- The value 1.96 (in R, `qnorm(0.975)` in the confidence interval is replaced by a new quantity, which we will denote as $t_{0.025}$
- The 95% confidence interval when σ is unknown is: $\bar{x} \pm t_{0.025} \times \frac{s}{\sqrt{n}}$
- 95% limits, upper and lower: `qt(0.975,df=24)`; `qt(0.025,df=24)` which give: $2.063899, -2.063899$.
- 90% limits, upper and lower: `qt(0.95,df=24)`; `qt(0.05,df=24)` which give: $1.710882, -1.710882$.
- (Note that R commands can be on the same line, if there is a semi-colon between them.)

Confidence Intervals when the Data is Normally Distributed (contd.)

- 95% confidence limits:
- `mean(x2) + qt(0.975,df=24) * sd(x2)/sqrt(25)`
- `mean(x2) - qt(0.975,df=24) * sd(x2)/sqrt(25)`
- Get: [247.9497, 339.7303] (better expressed as: [247.9, 339.7])
- 90% confidence limits:
- `mean(x2) + qt(0.95,df=24) * sd(x2)/sqrt(25)`
- `mean(x2) - qt(0.95,df=24) * sd(x2)/sqrt(25)`
- Get: [255.7989, 331.8811] (better expressed as: [255.8, 331.9])
- See how the latter, the 90% confidence interval, is tighter. We can say that the CI is broader to allow for 95% of the values, rather than just 90%, to lie in the range. Or we can say say that we can be 95% certain that an unknown population mean value lies somewhere in this range of values.

Confidence Intervals when the Data is Normally Distributed (contd.)

- Another note: 90% confidence interval, $\alpha = 0.10$.

Determination of Sample Size for Estimating a Population Mean

- Example 1
- A producer has introduced a new type of batteries for which the lifetime in hours is normally distributed. A sample yields the lifetimes data (in hours)
- 237, 242, 262, 244, 225, 218, 242, 248, 243, 234, 236, 228, 232, 254, 225
- Suppose it turns out that the standard deviation of the lifetime is 12 hours.
- Compute the sample size needed to obtain a 95% confidence interval for the expected lifetime with an interval width of one hour.

Determination of Sample Size for Estimating a Population Mean

- Example 1 (contd.)
- $\epsilon = z_\alpha \sigma / \sqrt{n} \Longrightarrow n = (z_\alpha \sigma / \epsilon)^2$
- $\sigma = 12$
- Just to check:
- `sd(c(237, 242, 262, 244, 225, 218, 242, 248, 243, 234, 236, 228, 232, 254, 225))`

  11.7108
- $n = 15$

- qnorm(0.95)

  [1] 1.644854

  qnorm(0.05)

  [1] −1.644854

  qnorm(0.95), multiply by 12, divide that by 1, take the square

  [1] 389.5983

- Answer: 390

Determination of Sample Size for Estimating a Population Mean

- Example 2
- The weight of eggs sold by the Midlands grocery store is normally distributed with a standard deviation of eight grams. Compute the sample size needed to obtain a 99% confidence interval for the expected eggs' weight with an interval width of at most two grams.
- $\epsilon = z_\alpha \sigma / \sqrt{n} \Longrightarrow n = (z_\alpha \sigma / \epsilon)^2$
- $\sigma = 8, \text{qnorm}(0.99) = 2.326348, \epsilon = 2$
- `qnorm(0.99) * 8 / 2`
  [1] 9.305391
- Answer: 10.

Another question

A random sample of 25 children selected from a large population had an average mark of 52 in a National examination. If the percentage marks can be assumed to follow a normal distribution with a standard deviation of 10, calculate 95% and 99% confidence intervals for the population average examination mark.

Another question

An electrical firm manufactures components that have a breaking strength that is approximately normally distributed with a standard deviation of 40 kg weight. If a random sample of 40 components has an average strength of 790 kg weight, find 95% and 98% confidence intervals for the population mean.

Estimation using Small Samples – using Gosset's t-distribution

Check the lecture notes. For small random samples from a normal population with unknown mean, μ, the distribution of the statistic $\frac{\bar{x}-\mu}{s/\sqrt{n}}$ follows a t distribution with $n-1$ degrees of freedom. A two-sided CI (confidence interval) can be found by using the critical values of the t distribution with $n-1$ degrees of freedom at the appropriate level. I.e. a two-sided $(100-\alpha)\%$ confidence interval is given by: $\bar{x} \pm t_{(n-1),\alpha/2} \ \frac{s}{\sqrt{n}}$.

Summary of Class 3, Lab Session 3

- Point estimation and interval estimation for population standard deviation known, and not known. (Former: standard normal distribution used; latter: Students t-distribution used.)